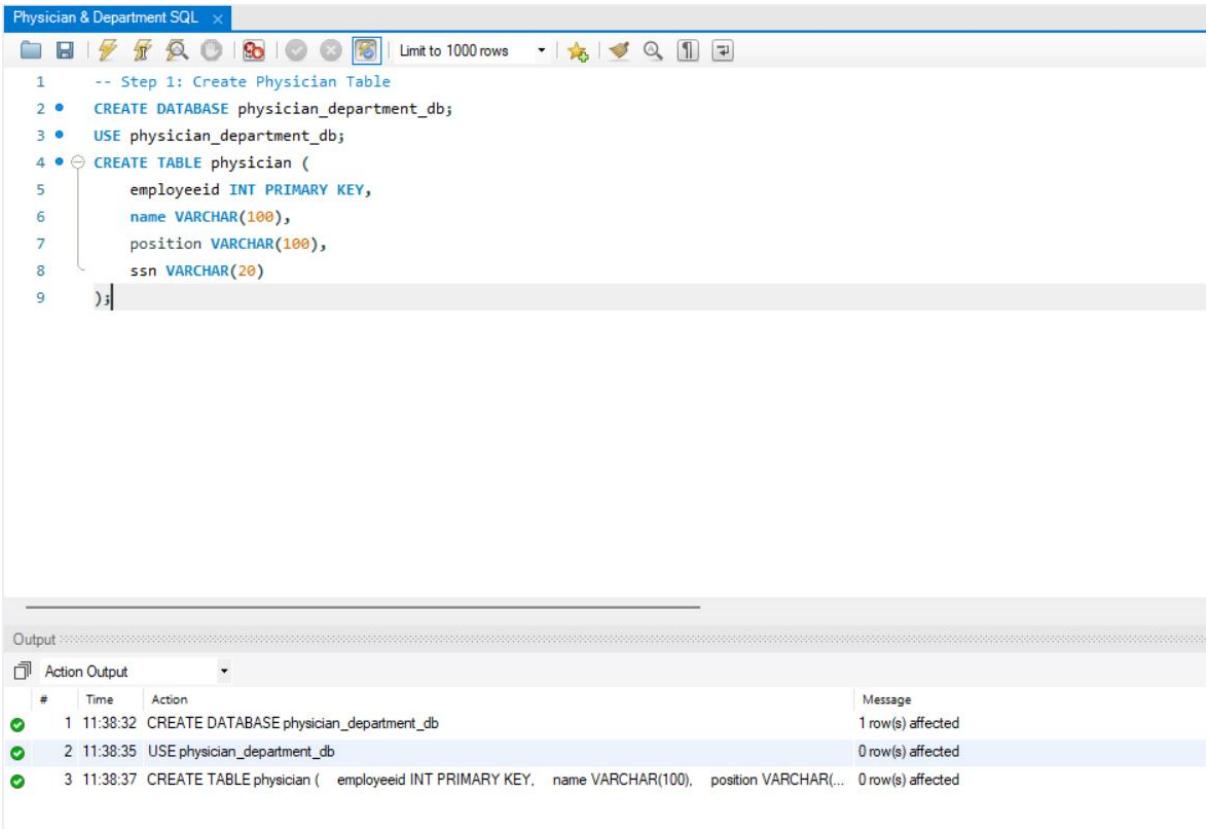
PHYSICIAN & DEPARTMENT SQL

Ili.Harshika (2520030607)

Question :

Perform SQL operations on Physician and Department tables using INNER JOIN, WHERE, LIKE, NOT LIKE, IN, NOT IN, EXISTS, GROUP BY, HAVING, ORDER BY, subqueries, COUNT(), COUNT(DISTINCT), MIN(), and MAX() to retrieve and analyse physician and department information.

Step 1 : Create Physician Database and Table



Physician & Department SQL

```sql
CREATE DATABASE physician_department_db;
USE physician_department_db;
CREATE TABLE physician (
    employeeid INT PRIMARY KEY,
    name VARCHAR(100),
    position VARCHAR(100),
    ssn VARCHAR(20)
);
-- Step 2: Insert Data into Physician
INSERT INTO physician VALUES
(1,'Harshika','Staff Internist','111111111'),
(2,'Ravinder','Attending Physician','222222222'),
(3,'Srilatha','Surgical Attending Physician','333333333'),
(4,'Santhvin','Senior Attending Physician','444444444'),
(5,'Varshika','Head Chief of Medicine','555555555'),
(6,'Manu','Surgical Attending Physician','666666666'),
(7,'Shiny','Surgical Attending Physician','777777777'),
(8,'Pranitha','MD Resident','888888888'),
(9,'Angela','Attending Psychiatrist','999999999');
```

Output

Action Output

#	Time	Action	Message
1	11:38:32	CREATE DATABASE physician_department_db	1 row(s) affected
2	11:38:35	USE physician_department_db	0 row(s) affected
3	11:38:37	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected
4	11:41:45	INSERT INTO physician VALUES (1,'Harshika','Staff Internist','111111111'), (2,'Ravinder','Attending Physician','2...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0

Physician & Department SQL

```sql
    ssn VARCHAR(20)
);
-- Step 2: Insert Data into Physician
INSERT INTO physician VALUES
(1,'Harshika','Staff Internist','111111111'),
(2,'Ravinder','Attending Physician','222222222'),
(3,'Srilatha','Surgical Attending Physician','333333333'),
(4,'Santhvin','Senior Attending Physician','444444444'),
(5,'Varshika','Head Chief of Medicine','555555555'),
(6,'Manu','Surgical Attending Physician','666666666'),
(7,'Shiny','Surgical Attending Physician','777777777'),
(8,'Pranitha','MD Resident','888888888'),
(9,'Angela','Attending Psychiatrist','999999999');
-- Step 3: Create Department Table
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);
```

Output

Action Output

#	Time	Action	Message
1	11:38:32	CREATE DATABASE physician_department_db	1 row(s) affected
2	11:38:35	USE physician_department_db	0 row(s) affected
3	11:38:37	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected
4	11:41:45	INSERT INTO physician VALUES (1,'Harshika','Staff Internist','111111111'), (2,'Ravinder','Attending Physician','2...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
5	11:42:13	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected

Physician & Department SQL

```
(2,'Ravinder','Attending Physician','222222222'),
(3,'Srilatha','Surgical Attending Physician','333333333'),
(4,'Santhvin','Senior Attending Physician','444444444'),
(5,'Varshika','Head Chief of Medicine','555555555'),
(6,'Manu','Surgical Attending Physician','666666666'),
(7,'Shiny','Surgical Attending Physician','777777777'),
(8,'Pranitha','MD Resident','888888888'),
(9,'Angela','Attending Psychiatrist','999999999');
-- Step 3: Create Department Table
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);
-- Step 4: Insert Data into Department
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
```

Output

Action Output

#	Time	Action	Message
1	11:38:32	CREATE DATABASE physician_department_db	1 row(s) affected
2	11:38:35	USE physician_department_db	0 row(s) affected
3	11:38:37	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(...	0 row(s) affected
4	11:41:45	INSERT INTO physician VALUES (1,'Harshika','Staff Internist','111111111'), (2,'Ravinder','Attending Physician','2...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
5	11:42:13	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
6	11:43:08	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Step 2 : JOIN

Physician & Department SQL

```
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
-- Step 5: INNER JOIN
SELECT d.name AS Department,
       p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department	Head_Physician
General Medicine	Santhvin
Surgery	Shiny
Psychiatry	Angela

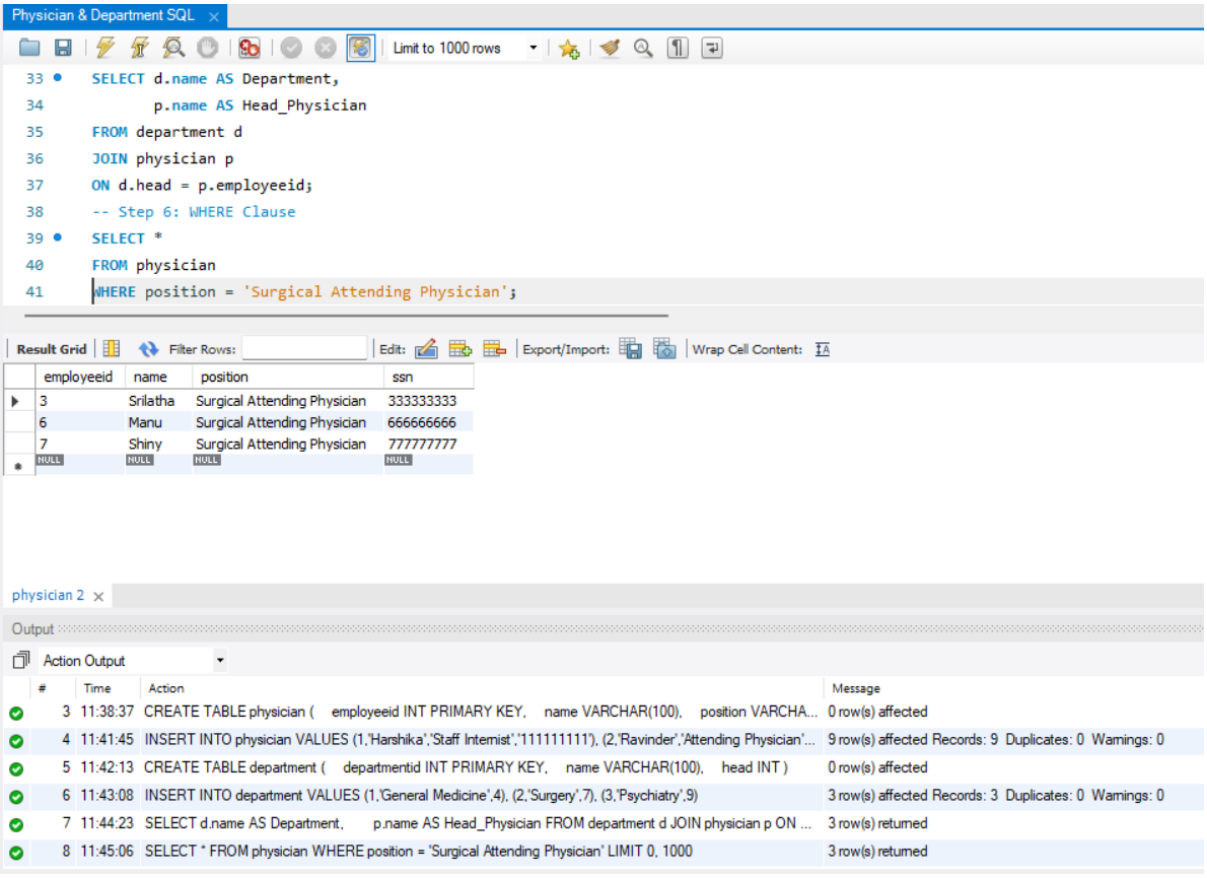
Result 1

Output

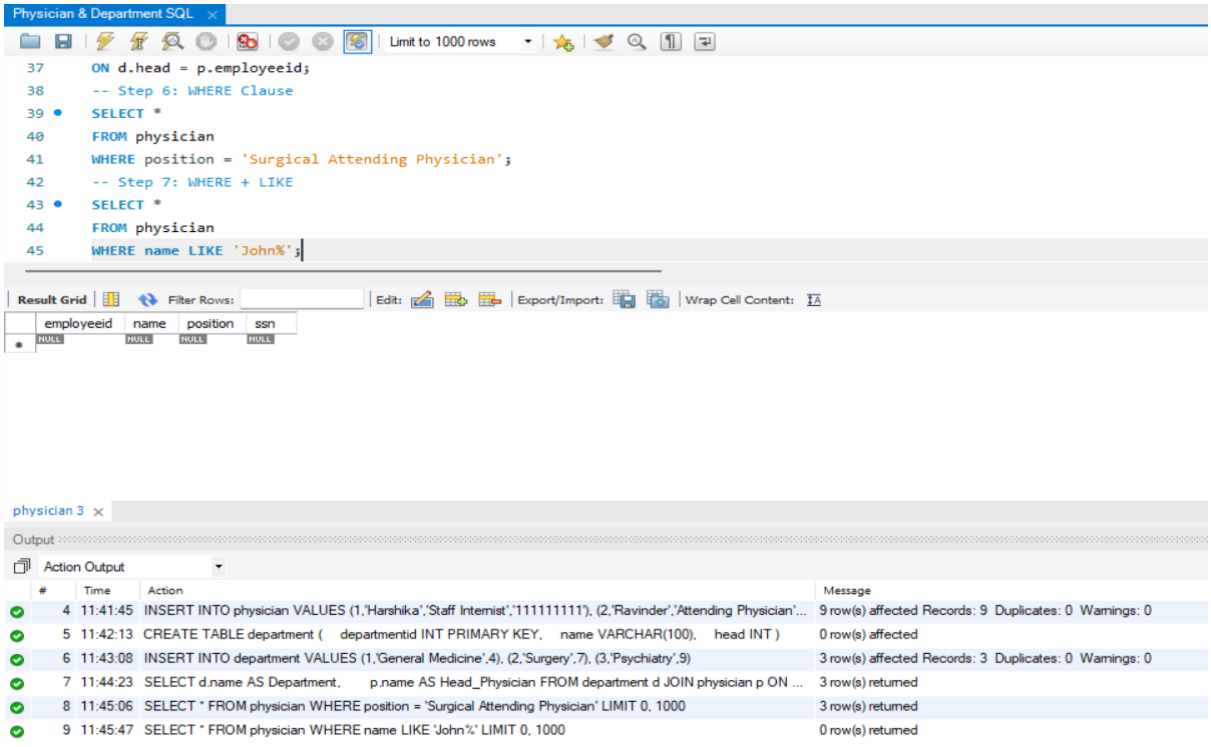
Action Output

#	Time	Action	Message
2	11:38:35	USE physician_department_db	0 row(s) affected
3	11:38:37	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA...	0 row(s) affected
4	11:41:45	INSERT INTO physician VALUES (1,'Harshika','Staff Internist','111111111'), (2,'Ravinder','Attending Physician'...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
5	11:42:13	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
6	11:43:08	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7	11:44:23	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned

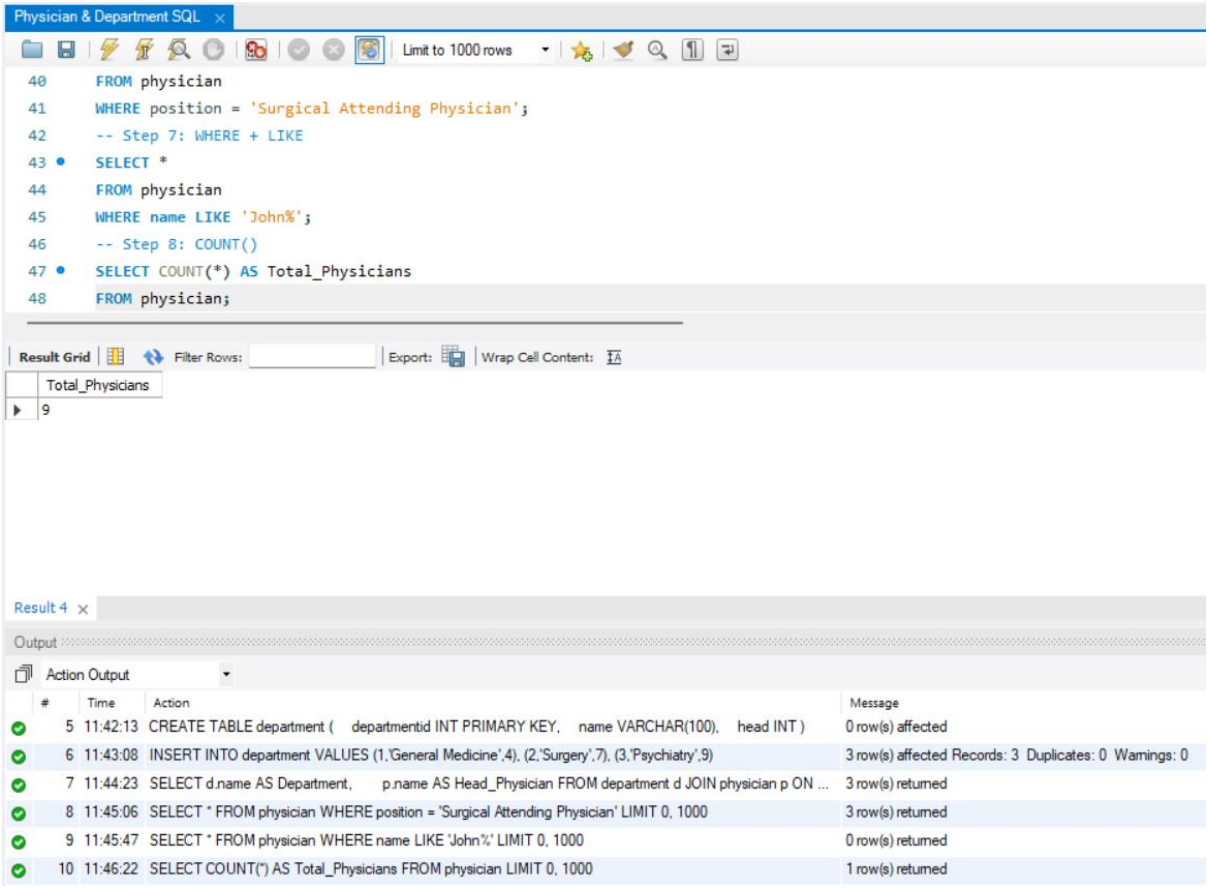
Step 3 : WHERE

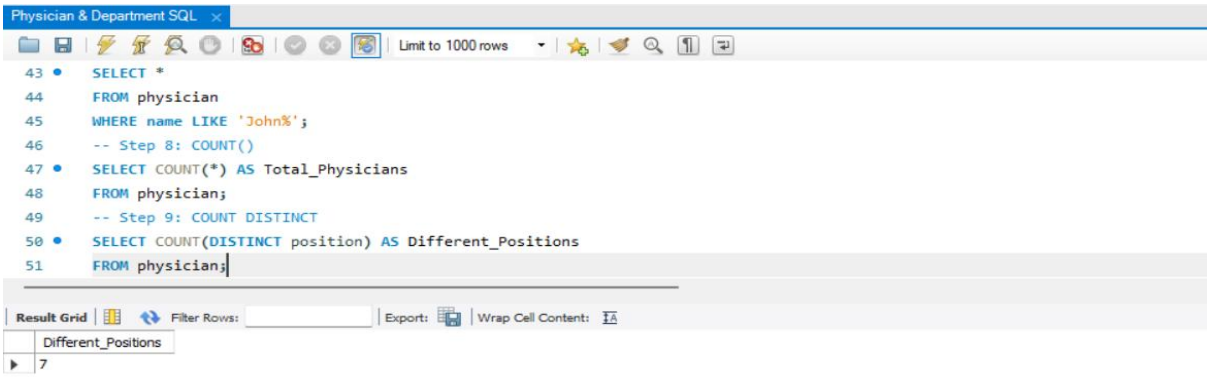


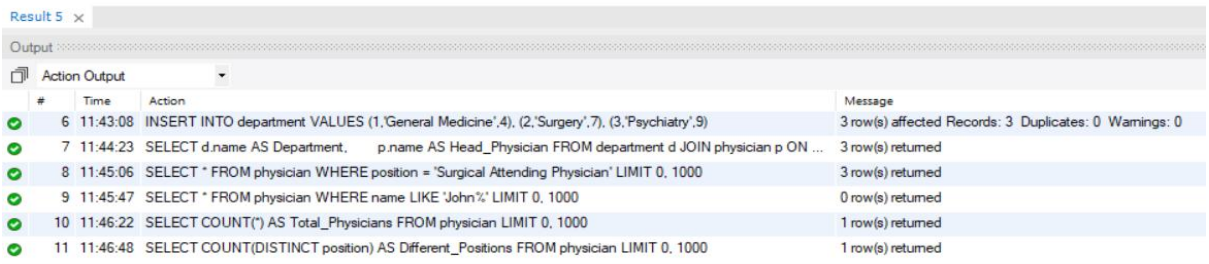
Step 4 : LIKE



Step 5 : COUNT


Physician & Department SQL
Limit to 1000 rows
FROM physician
WHERE position = 'Surgical Attending Physician';
-- Step 7: WHERE + LIKE
SELECT *
FROM physician
WHERE name LIKE 'John%';
-- Step 8: COUNT()
SELECT COUNT(*) AS Total_Physicians
FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Total_Physicians
9
Result 4
Output
Action Output
Time Action Message
5 11:42:13 CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT) 0 row(s) affected
6 11:43:08 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7 11:44:23 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... 3 row(s) returned
8 11:45:06 SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000 3 row(s) returned
9 11:45:47 SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000 0 row(s) returned
10 11:46:22 SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000 1 row(s) returned


Physician & Department SQL
Limit to 1000 rows
SELECT *
FROM physician
WHERE name LIKE 'John%';
-- Step 8: COUNT()
SELECT COUNT(*) AS Total_Physicians
FROM physician;
-- Step 9: COUNT DISTINCT
SELECT COUNT(DISTINCT position) AS Different_Positions
FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Different_Positions
7


Result 5
Output
Action Output
Time Action Message
6 11:43:08 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7 11:44:23 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... 3 row(s) returned
8 11:45:06 SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000 3 row(s) returned
9 11:45:47 SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000 0 row(s) returned
10 11:46:22 SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000 1 row(s) returned
11 11:46:48 SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000 1 row(s) returned

Step 6 : GROUP BY

Physician & Department SQL

```
48      FROM physician;
49      -- Step 9: COUNT DISTINCT
50      SELECT COUNT(DISTINCT position) AS Different_Positions
51      FROM physician;
52      -- Step 10: GROUP BY
53      SELECT position,
54             COUNT(*) AS Total_Employees
55      FROM physician
56      GROUP BY position;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 6

Output

Action Output

#	Time	Action	Message
7	11:44:23	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned

Physician & Department SQL

```
54              COUNT(*) AS Total_Employees
55      FROM physician
56      GROUP BY position;
57      -- Step 11: GROUP BY + HAVING
58      SELECT position,
59             COUNT(*) AS Total_Employees
60      FROM physician
61      GROUP BY position
62      HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

position	Total_Employees
Surgical Attending Physician	3

Result 7

Output

Action Output

#	Time	Action	Message
8	11:45:06	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
13	11:47:59	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned

Step 7: ORDER BY

Physician & Department SQL

```
58  SELECT position,
59         COUNT(*) AS Total_Employees
60  FROM physician
61  GROUP BY position
62  HAVING COUNT(*) > 1;
63  -- Step 12: ORDER BY ASC
64  SELECT *
65  FROM physician
66  ORDER BY name ASC;
```

Result Grid

employeeid	name	position	ssn
9	Angela	Attending Psychiatrist	999999999
1	Harshika	Staff Internist	111111111
6	Manu	Surgical Attending Physician	666666666
8	Pranitha	MD Resident	888888888
2	Ravinder	Attending Physician	222222222
4	Santhvin	Senior Attending Physician	444444444
7	Shiny	Surgical Attending Physician	777777777
3	Srilatha	Surgical Attending Physician	333333333
5	Varshika	Head Chief of Medicine	555555555

physician 8

Output

Action Output

#	Time	Action	Message
9	11:45:47	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
13	11:47:59	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
14	11:48:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned

Physician & Department SQL

```
62  HAVING COUNT(*) > 1;
63  -- Step 12: ORDER BY ASC
64  SELECT *
65  FROM physician
66  ORDER BY name ASC;
67  -- Step 13: ORDER BY DESC
68  SELECT *
69  FROM physician
70  ORDER BY employeeid DESC;
```

Result Grid

employeeid	name	position	ssn
9	Angela	Attending Psychiatrist	999999999
8	Pranitha	MD Resident	888888888
7	Shiny	Surgical Attending Physician	777777777
6	Manu	Surgical Attending Physician	666666666
5	Varshika	Head Chief of Medicine	555555555
4	Santhvin	Senior Attending Physician	444444444
3	Srilatha	Surgical Attending Physician	333333333
2	Ravinder	Attending Physician	222222222
1	Harshika	Staff Internist	111111111

physician 9

Output

Action Output

#	Time	Action	Message
10	11:46:22	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
13	11:47:59	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
14	11:48:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
15	11:49:03	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned

Step 8 : IN

Physician & Department SQL

```
70      ORDER BY employeeid DESC;
71      -- Step 14: WHERE + IN
72  •   SELECT *
73      FROM physician
74      WHERE employeeid IN
75      (
76          SELECT head
77          FROM department
78      );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employeeid	name	position	ssn
4	Santhvin	Senior Attending Physician	444444444
7	Shiny	Surgical Attending Physician	777777777
9	Angela	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

physician 10

Output

Action Output

#	Time	Action	Message
11	11:46:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
13	11:47:59	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
14	11:48:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
15	11:49:03	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
16	11:53:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned

Step 9 : NOT IN

Physician & Department SQL

```
78      );
79      -- Step 15: NOT IN
80  •   SELECT *
81      FROM physician
82      WHERE employeeid NOT IN
83      (
84          SELECT head
85          FROM department
86      );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employeeid	name	position	ssn
1	Harshika	Staff Internist	111111111
2	Ravinder	Attending Physician	222222222
3	Srilatha	Surgical Attending Physician	333333333
5	Varshika	Head Chief of Medicine	555555555
6	Manu	Surgical Attending Physician	666666666
8	Pranitha	MD Resident	888888888
NULL	NULL	NULL	NULL

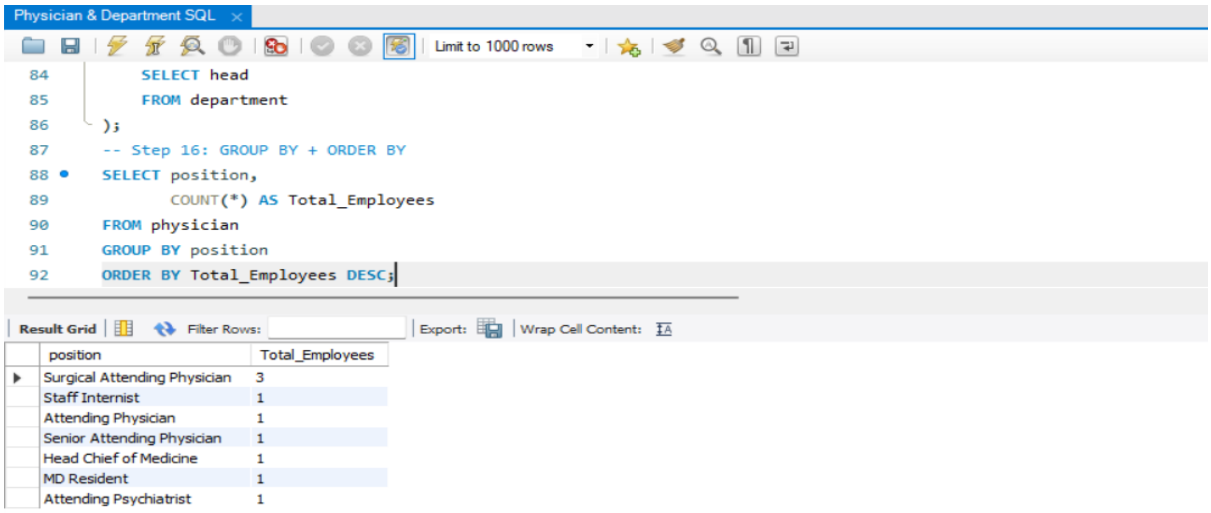
physician 11

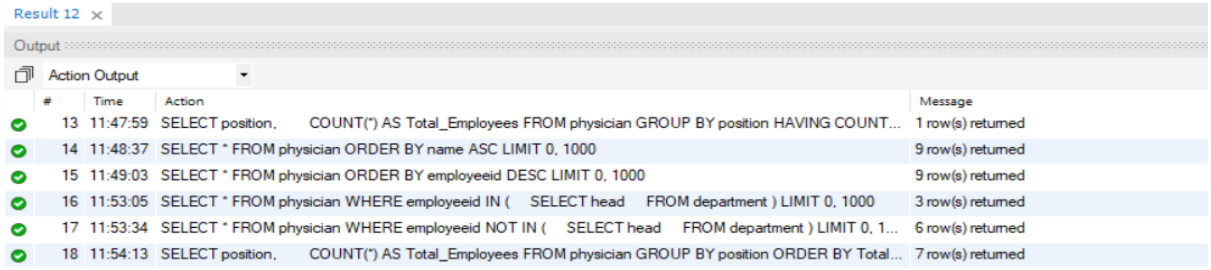
Output

Action Output

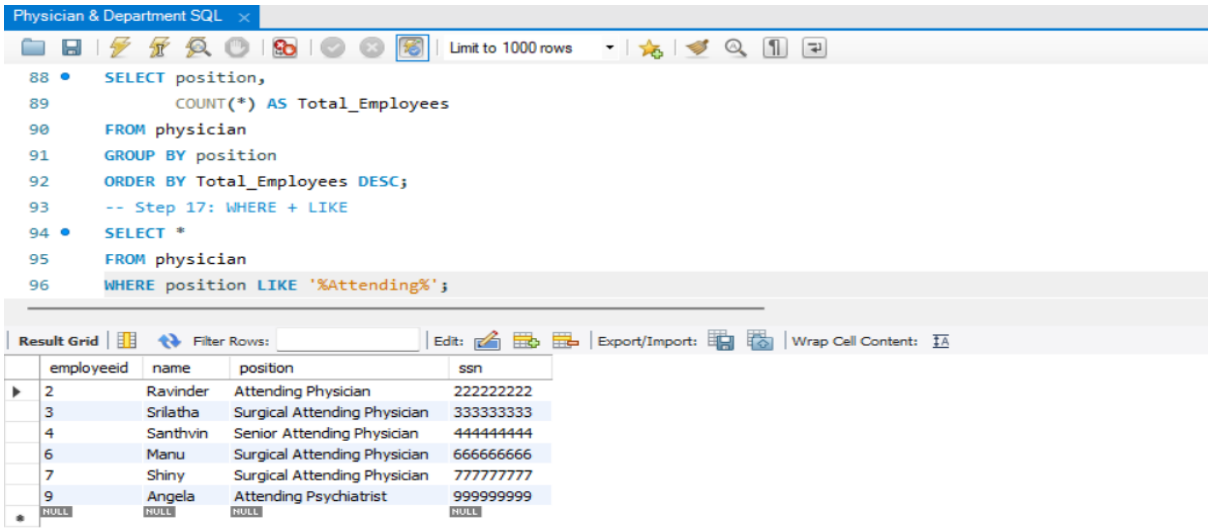
#	Time	Action	Message
12	11:47:37	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
13	11:47:59	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
14	11:48:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
15	11:49:03	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
16	11:53:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
17	11:53:34	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned

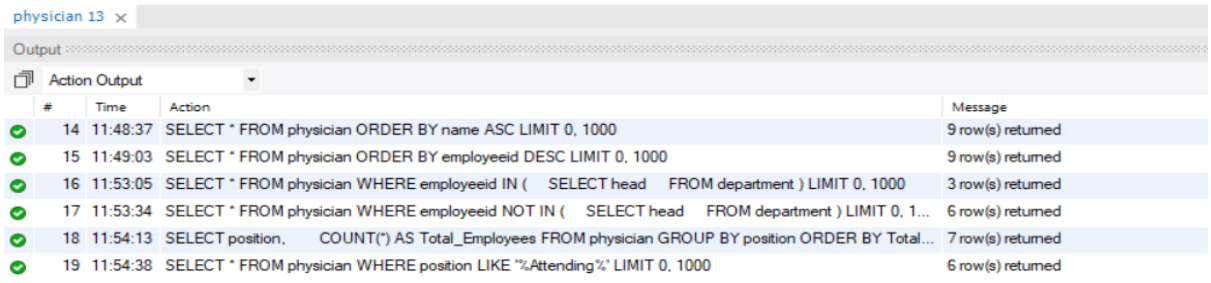
Step 10 : GROUP BY + ORDER BY



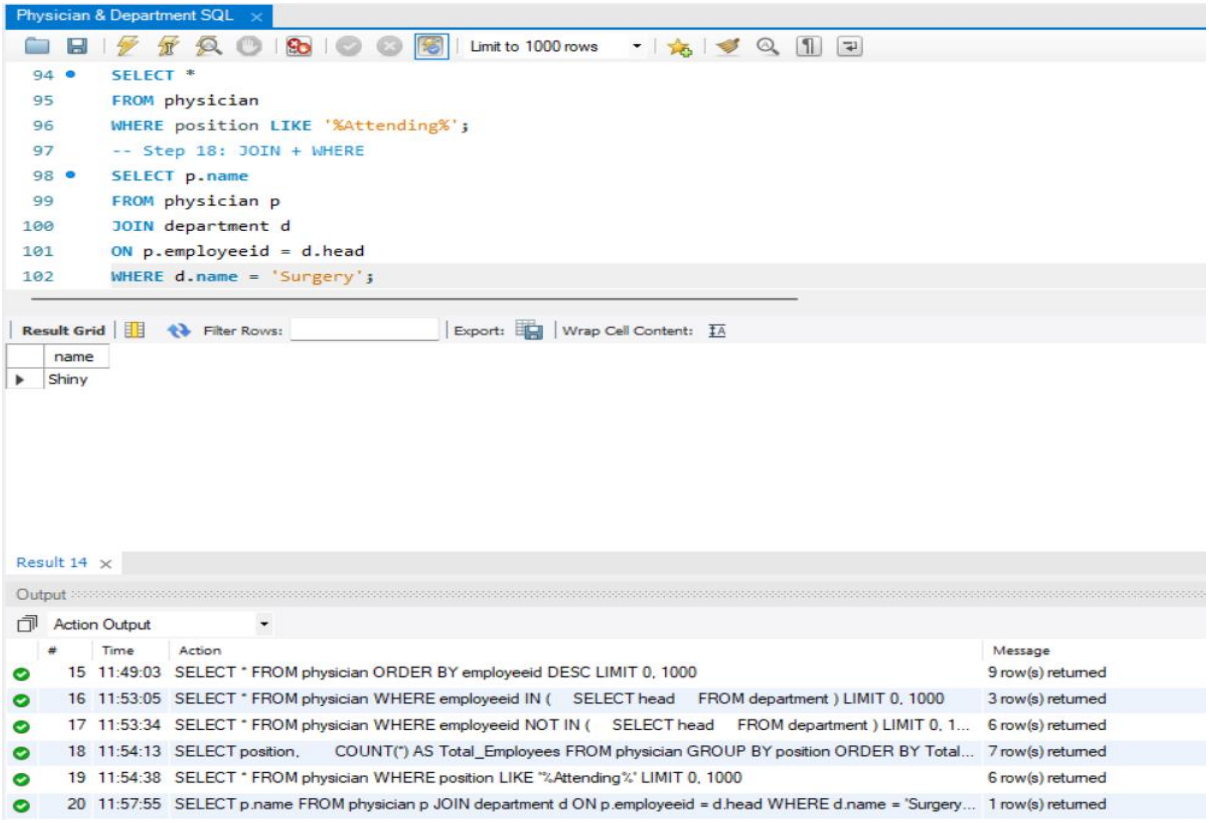


Step 11: LIKE

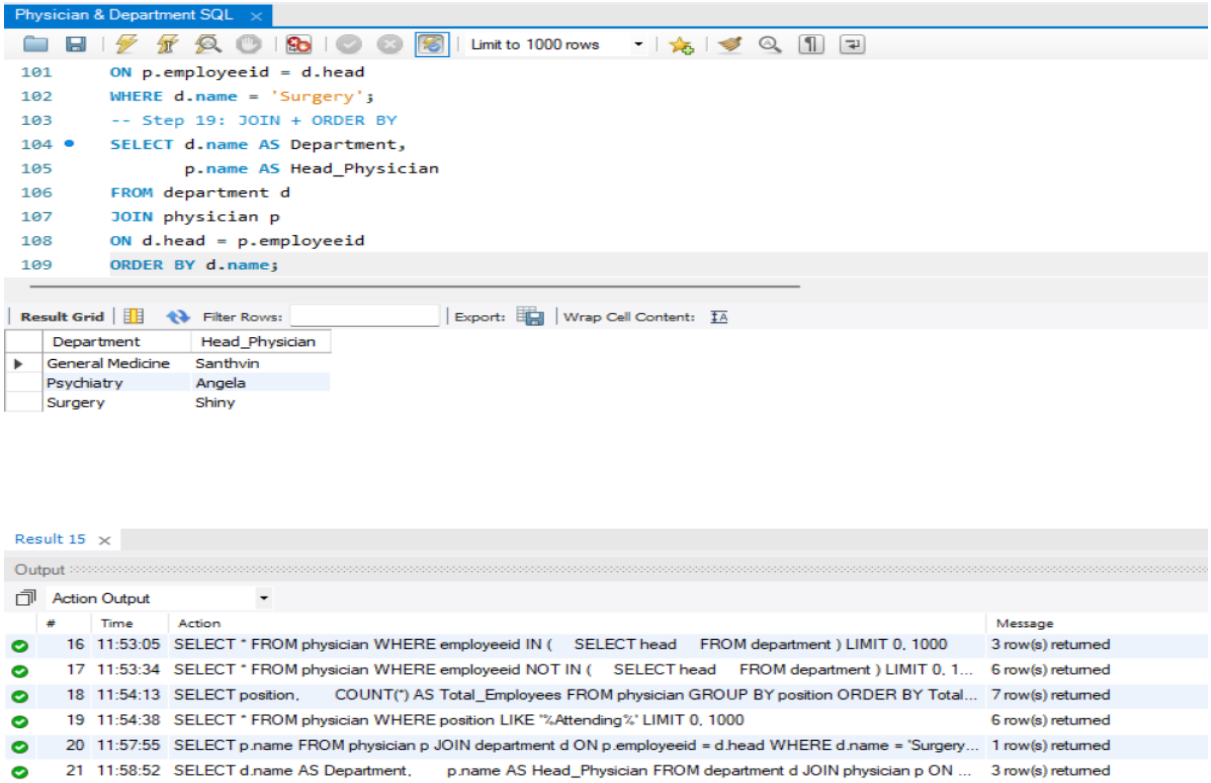




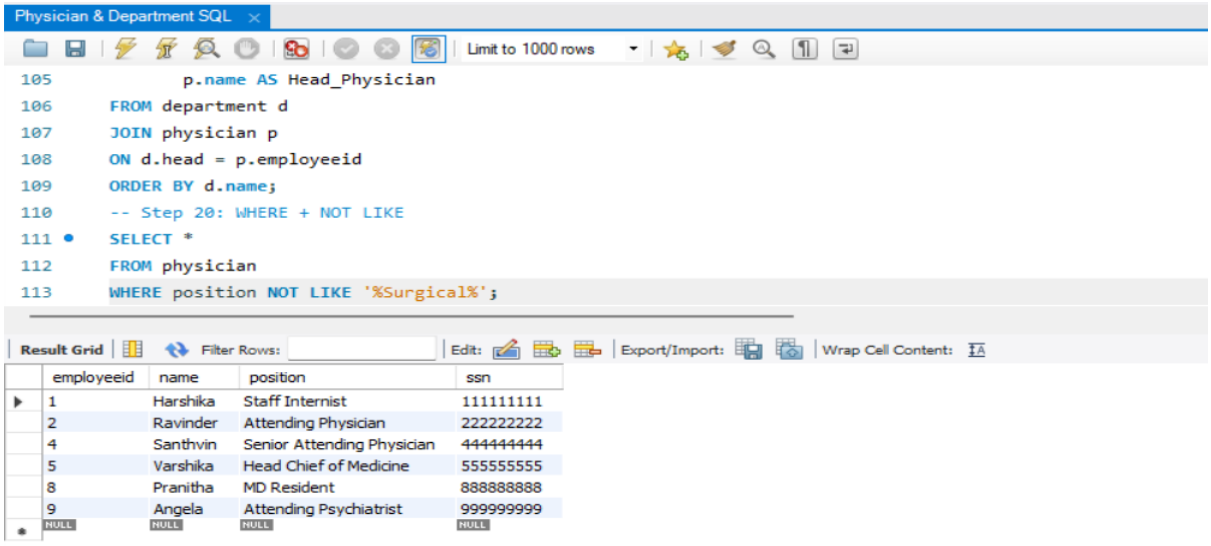
Step 12: JOIN + WHERE



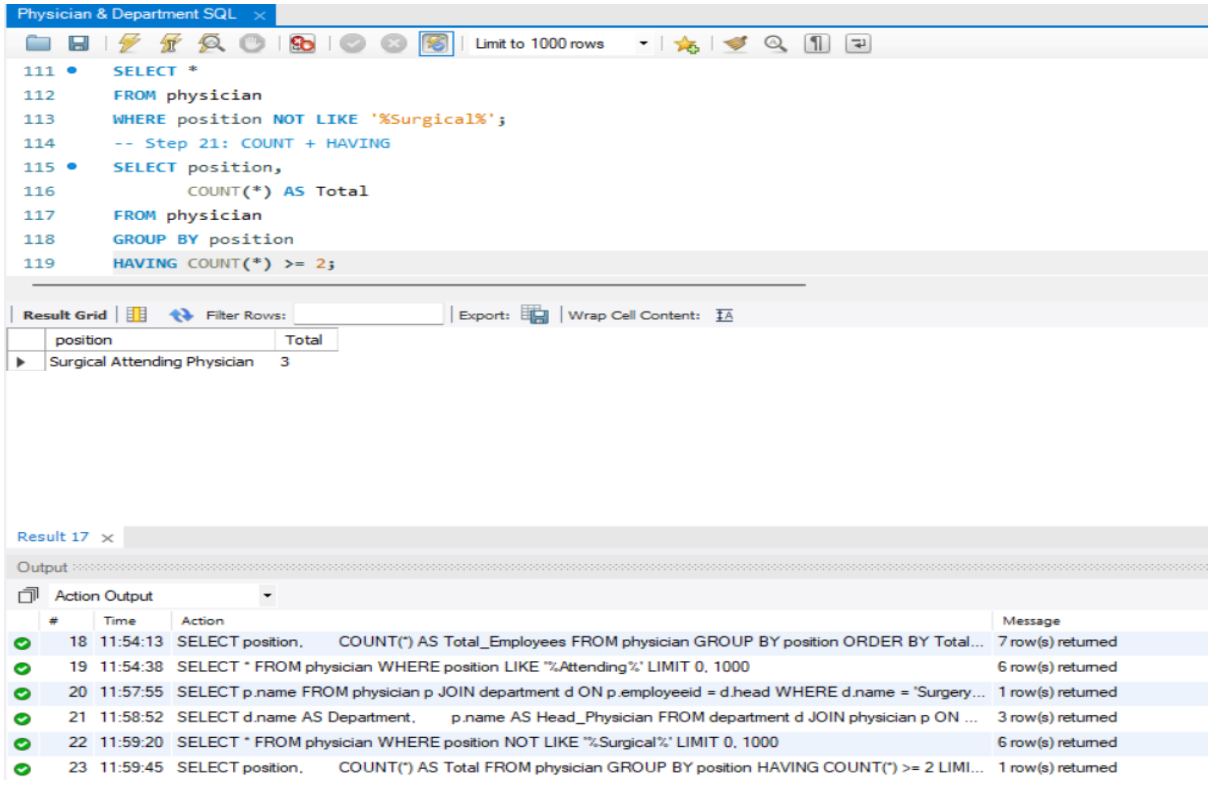
Step 13: JOIN+ORDER BY



Step 14: NOT LIKE



Step 15: COUNT+HAVING



Step 16: EXISTS

Physician & Department SQL

Limit to 1000 rows

```
120     -- Step 22: EXISTS
121 •   SELECT *
122     FROM department d
123     WHERE EXISTS
124     (
125         SELECT 1
126         FROM physician p
127         WHERE p.employeeid = d.head
128     );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

department 18

Output

Action Output

#	Time	Action	Message
19	11:54:38	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
20	11:57:55	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMI...	1 row(s) returned
24	12:00:12	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned

Step 17: MAX

Physician & Department SQL

Limit to 1000 rows

```
123     WHERE EXISTS
124     (
125         SELECT 1
126         FROM physician p
127         WHERE p.employeeid = d.head
128     );
129     -- Step 23: MAX()
130 •   SELECT MAX(employeeid) AS Highest_Employee_ID
131     FROM physician;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Highest_Employee_ID
9

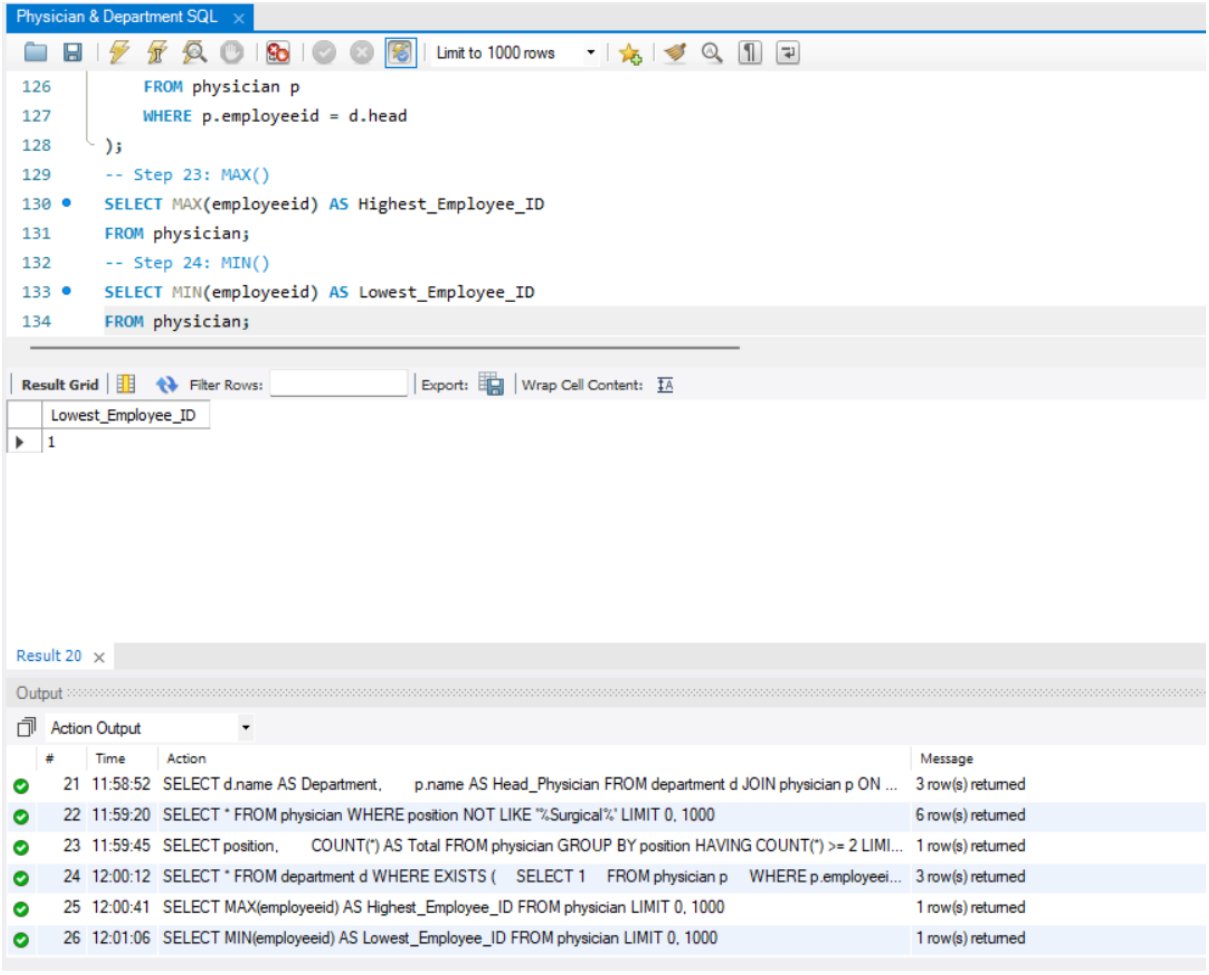
Result 19

Output

Action Output

#	Time	Action	Message
20	11:57:55	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
21	11:58:52	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
22	11:59:20	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
23	11:59:45	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMI...	1 row(s) returned
24	12:00:12	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned
25	12:00:41	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned

Step 18: MIN


Physician & Department SQL
Limit to 1000 rows
126 FROM physician p
127 WHERE p.employeeid = d.head
128);
129 -- Step 23: MAX()
130 SELECT MAX(employeeid) AS Highest_Employee_ID
131 FROM physician;
132 -- Step 24: MIN()
133 SELECT MIN(employeeid) AS Lowest_Employee_ID
134 FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Lowest_Employee_ID
1
Result 20
Output
Action Output
Time Action Message
21 11:58:52 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... 3 row(s) returned
22 11:59:20 SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000 6 row(s) returned
23 11:59:45 SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMI... 1 row(s) returned
24 12:00:12 SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei... 3 row(s) returned
25 12:00:41 SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000 1 row(s) returned
26 12:01:06 SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LIMIT 0, 1000 1 row(s) returned